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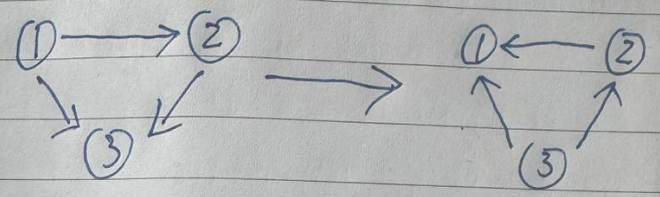
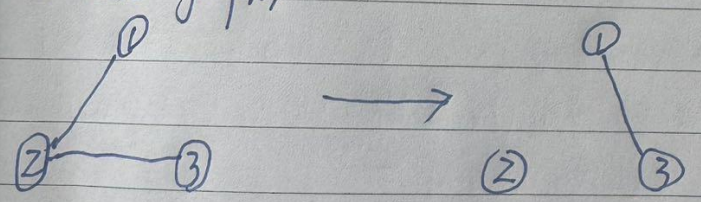
ex 7

Problem 1.

1. transposed graphs



2. inverse graphs



3. Q: what happens if the original is dense for the inverse?

A: The more dense original is, ~~the~~ more sparse the ~~of~~ inverse graph will be.

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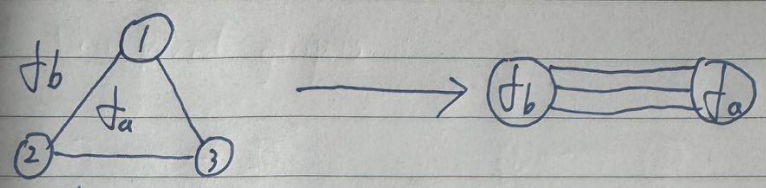
4. dual graph.

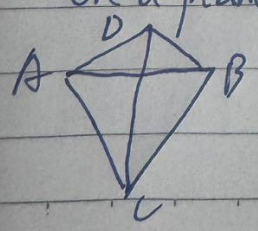


f_a: inside

f_b: out side

5. Q: Why is the dual only well-defined for planner graphs? Provide an example of a non-planar graph and explain why it has no dual.

A: The dual graph is constructed by placing a vertex inside each face of the graph, and connecting two vertices whenerer their corresponding face share an edge. This construction requires the graph to have well-defind face, which ~~can~~ ~~be~~ only exist when the graph can be drawn on a plane with no edge crossings.



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Problem 2.

1. initial call

$R = \{\}$, $P = \{A, B, C, D\}$, $X = \{\}$

2. Trace of the first two recursive calls

Call 1. pick $v = A$ from P

	value
R	$\{A\}$
P	$P \cap N(A) = \{A, B, C, D\} \cap \{B, C\} = \{B, C\}$
X	$X \cap N(A) = \{\} \cap \{B, C\} = \{\}$

→ Recurse: Bronkerbosch($\{A\}$, $\{B, C\}$, $\{\}$)

Call 2. pick $v = B$ from $P = \{B, C\}$

	value
R	$\{A, B\}$
P	$P \cap N(B) = \{B, C\} \cap \{A, C\} = \{C\}$
X	$X \cap N(B) = \{\} \cap \{A, C\} = \{\}$

→ Recurse: Bronkerbosch($\{A, B\}$, $\{C\}$, $\{\}$)

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Call 3. pick $v = C$ from $P = \{C\}$

	value
R	$\{A, B, C\}$
P	$P \cap N(C) = \{C\} \cap \{A, B, D\} = \{\}$
X	$X \cap N(C) = \{\} \cap \{A, B, D\} = \{\}$

→ Recurse: Bronkerbosch($\{A, B, C\}$, $\{\}$, $\{\}$)

find the first maxium clique: $\{A, B, C\}$

after backtracking we can find the second $\{C, D\}$

3. All maximal cliques – which is the maximum?

clique	size	type
$\{A, B, C\}$	3	maxima and maxium
$\{C, D\}$	2	maxima.